Week 7 Tutorial Solutions

Question 28

Evaluate $\displaystyle\int_0^5 \frac{1}{(x+1)^3}\,dx$.

Find an antiderivative $F(x)$ by guess and check. Recall

$$\int_a^b f(x)\,dx = F(b) - F(a).$$

Take

$$F(x) = \frac{-1}{2(x+1)^2}.$$

Check:

$$F'(x) = -\frac{1}{2}\cdot(-2)\,(x+1)^{-3}\cdot(1+0) = \frac{1}{(x+1)^3}.$$

Therefore

$$\begin{aligned}\int_0^5 \frac{1}{(x+1)^3}\,dx &= \left[\frac{-1}{2(x+1)^2}\right]_0^5 \\ &= \frac{-1}{2(6)^2} - \left(\frac{-1}{2(1)^2}\right) \\ &= -\frac{1}{72} + \frac{1}{2} \\ &= \boxed{\frac{35}{72}}\end{aligned}$$

Sneak Peek: Solving by Substitution

Let $u = x + 1$. Then $du = dx$.
Change the bounds:

$$x = 0 \implies u = 1, \qquad x = 5 \implies u = 6.$$

Then

$$\begin{aligned}
\int_0^5 \frac{1}{(x+1)^3}\,dx &= \int_1^6 u^{-3}\,du \\
&= \left[\frac{u^{-2}}{-2}\right]_1^6 \\
&= \left[\frac{-1}{2u^2}\right]_1^6 \\
&= \frac{-1}{2(6)^2} - \left(\frac{-1}{2(1)^2}\right) \\
&= -\frac{1}{72} + \frac{1}{2} \\
&= \boxed{\frac{35}{72}}
\end{aligned}$$

Same answer as guess and check, but with no guessing needed.

Question 29

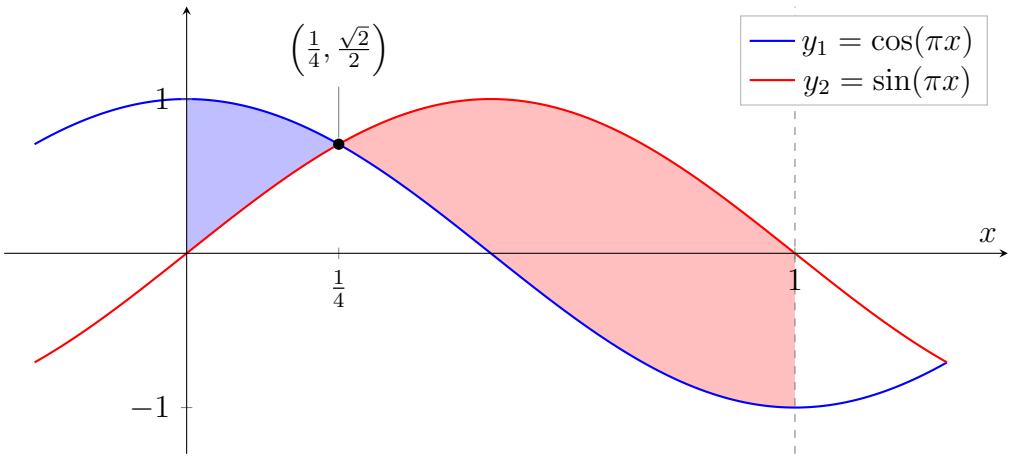
Consider $y_1 = \cos(\pi x)$ and $y_2 = \sin(\pi x)$.

(a)

Area between curves = Area top − Area bottom.

The curves intersect when $\cos(\pi x) = \sin(\pi x)$, that is, when $\pi x = \frac{\pi}{4}$, so $x = \frac{1}{4}$. This splits $[0, 1]$ into two sections: $\left[0, \frac{1}{4}\right]$ and $\left[\frac{1}{4}, 1\right]$.

- On $\left[0, \frac{1}{4}\right]$, $\cos(\pi x)$ is above $\sin(\pi x)$.
- On $\left[\frac{1}{4}, 1\right]$, $\sin(\pi x)$ is above $\cos(\pi x)$.



Only the shaded region over $[0, 1]$ is counted. Therefore

$$\text{Area} = \text{Blue Area} + \text{Red Area}$$

$$\text{Area} = \int_0^{1/4} \big(\cos(\pi x) - \sin(\pi x)\big)\, dx + \int_{1/4}^{1} \big(\sin(\pi x) - \cos(\pi x)\big)\, dx.$$

(b)

First integral:

$$\begin{aligned}\int_0^{1/4} \big(\cos(\pi x) - \sin(\pi x)\big)\, dx &= \left[\frac{1}{\pi}\big(\sin(\pi x) + \cos(\pi x)\big)\right]_0^{1/4} \\ &= \frac{1}{\pi}\left(\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}\right) - \frac{1}{\pi}(0+1) \\ &= \frac{1}{\pi}\big(\sqrt{2}-1\big)\end{aligned}$$

Second integral:

$$\begin{aligned}\int_{1/4}^{1} \big(\sin(\pi x) - \cos(\pi x)\big)\, dx &= \frac{1}{\pi}\Big[-\cos(\pi x) - \sin(\pi x)\Big]_{1/4}^{1} \\ &= \frac{1}{\pi}\big(1-0\big) - \frac{1}{\pi}\left(-\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}\right) \\ &= \frac{1}{\pi}\big(\sqrt{2}+1\big)\end{aligned}$$

Adding the two areas:

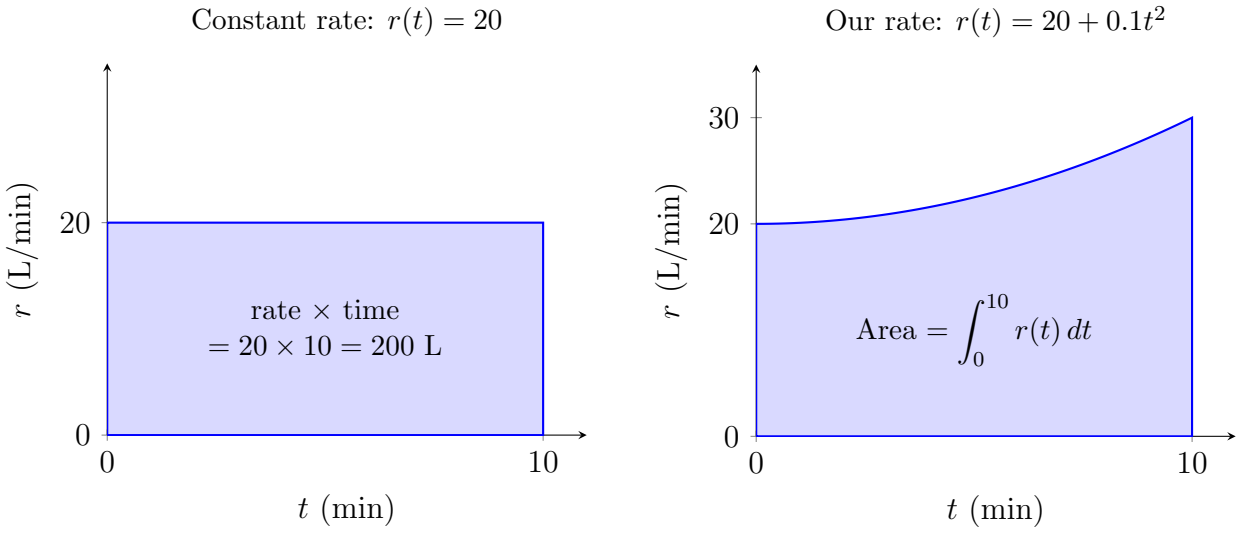
$$A = \frac{1}{\pi}\big(\sqrt{2}-1\big) + \frac{1}{\pi}\big(\sqrt{2}+1\big) = \boxed{\frac{2\sqrt{2}}{\pi}}$$

Question 30

A pump adds water to a tank at a rate of $r(t) = 20 + 0.1t^2$ L/min for $t \in [0, 10]$ minutes, and $V(0) = 200$ L. How much water is in the tank after 10 minutes?

Total volume = initial volume + total water added. The initial volume is known, so we need the total added.

Since $r(t)$ is measured in L/min, multiplying by time (min) gives litres. If the rate were constant, say $r(t) = 20$ L/min, the water added in 10 minutes would be rate $\times$ time $= 20 \times 10 = 200$ L, which is the area of a rectangle under the graph. Our rate changes with time, but the idea is the same: the water added is the area under the graph of $r(t)$ over $[0, 10]$.



So, with our bounds $t \in [0, 10]$:

$$\begin{aligned}
\text{Water added} &= \int_0^{10} \left(20 + 0.1t^2\right) dt \\
&= \left[20t + \frac{1}{3}(0.1)\, t^3\right]_0^{10} \\
&= \left(200 + \frac{1}{3}(0.1)(1000)\right) - 0 \\
&= 200 + \frac{100}{3} \approx 233.3 \text{ L}
\end{aligned}$$

Therefore

$$V_{\text{total}} = 200 \text{ L} + 233.3 \text{ L} = \boxed{433.3 \text{ L}}$$

Question 31

Recall the reverse power rule: for $n \neq -1$,

$$\int x^n \, dx = \frac{1}{n+1} x^{n+1} + C.$$

Add 1 to the power, then divide by the new power. Always simplify powers first.

(a)

$$\begin{aligned}\int \frac{1}{x^2} \, dx = \int x^{-2} \, dx &= \frac{1}{-2+1} x^{-2+1} + C \\ &= -1 \cdot x^{-1} + C \\ &= \boxed{-\frac{1}{x} + C}\end{aligned}$$

(b)

$$\begin{aligned}\int \frac{1}{\sqrt{x}} \, dx = \int x^{-1/2} \, dx &= \frac{1}{-\frac{1}{2}+1} x^{-\frac{1}{2}+1} + C \\ &= \frac{1}{(1/2)} x^{1/2} + C \\ &= \boxed{2\sqrt{x} + C}\end{aligned}$$

(c)

$$\begin{aligned}\int \sqrt{x} \, dx = \int x^{1/2} \, dx &= \frac{1}{\frac{1}{2}+1} x^{\frac{1}{2}+1} + C \\ &= \frac{1}{(3/2)} x^{3/2} + C \\ &= \boxed{\frac{2}{3} x^{3/2} + C}\end{aligned}$$

(d)

The reverse power rule does not work here, since it would require division by zero:

$$\int \frac{1}{x}\,dx = \int x^{-1}\,dx \neq \frac{1}{-1+1}\,x^{-1+1}.$$

Instead,

$$\boxed{\int \frac{1}{x}\,dx = \ln|x| + C}$$

Remember the absolute value sign!

Why $\ln|x|$? The easiest way to remember it is

$$\frac{d}{dx}\big[\ln(x)\big] = \frac{1}{x}.$$

Reversing this, an antiderivative of $\frac{1}{x}$ is $\ln(x)$, and that is the only function that works, since no power of x differentiates to x^{-1}.

Why the absolute value? The function $\frac{1}{x}$ is defined for both $x > 0$ and $x < 0$, but $\ln(x)$ is only defined for $x > 0$. Using $\ln|x|$ covers both cases:

- For $x > 0$: $|x| = x$, so $\dfrac{d}{dx}\ln|x| = \dfrac{d}{dx}\ln(x) = \dfrac{1}{x}$.
- For $x < 0$: $|x| = -x$, so by the chain rule, $\dfrac{d}{dx}\ln(-x) = \dfrac{1}{-x}\cdot(-1) = \dfrac{1}{x}$.

In both cases the derivative of $\ln|x|$ is $\frac{1}{x}$, so $\ln|x| + C$ is an antiderivative on the whole domain of $\frac{1}{x}$.

Similarities and Differences

Similarities:

- (a), (b), (c) are algebraic functions.
- (b), (c) are positive-valued for $x > 0$ (they are not defined for $x < 0$).
- All have $+C$.

Differences:

- (d) is a *logarithmic* function.

Question 32

Evaluate $\displaystyle\int \frac{x^3-2}{\sqrt{x}}\,dx$.

Rewrite the fraction:

$$\int \frac{x^3-2}{\sqrt{x}}\,dx = \int \frac{x^3-2}{x^{1/2}}\,dx = \int \left(\frac{x^3}{x^{1/2}} - \frac{2}{x^{1/2}}\right) dx = \int \left(x^{5/2} - 2x^{-1/2}\right) dx.$$

Use the reverse power rule as normal:

$$\int \left(x^{5/2} - 2x^{-1/2}\right) dx = \frac{1}{\frac{5}{2}+1}\, x^{\frac{5}{2}+1} - \frac{2}{-\frac{1}{2}+1}\, x^{-\frac{1}{2}+1} + C.$$

Simplify:

$$\boxed{\int \frac{x^3-2}{\sqrt{x}}\,dx = \frac{2}{7}\,x^{7/2} - 4\sqrt{x} + C}$$

Question 33

Given

$$\int_0^9 f(x)\,dx = 37, \qquad \int_0^9 g(x)\,dx = 16,$$

find $\displaystyle\int_0^9 \big[2f(x) + 3g(x)\big]\,dx$.

Properties of integration:

1. **Additivity:**
$$\int_a^b \big[f(x)+g(x)\big]\,dx = \int_a^b f(x)\,dx + \int_a^b g(x)\,dx$$
2. **Scaling** (α is a constant):
$$\int_a^b \alpha f(x)\,dx = \alpha \int_a^b f(x)\,dx$$

Recall these properties from differentiation!

Applying them:

$$\begin{aligned}\int_0^9 \big[2f(x) + 3g(x)\big]\,dx &= \int_0^9 2f(x)\,dx + \int_0^9 3g(x)\,dx \\ &= 2\int_0^9 f(x)\,dx + 3\int_0^9 g(x)\,dx \\ &= 2(37) + 3(16) \\ &= \boxed{122}\end{aligned}$$